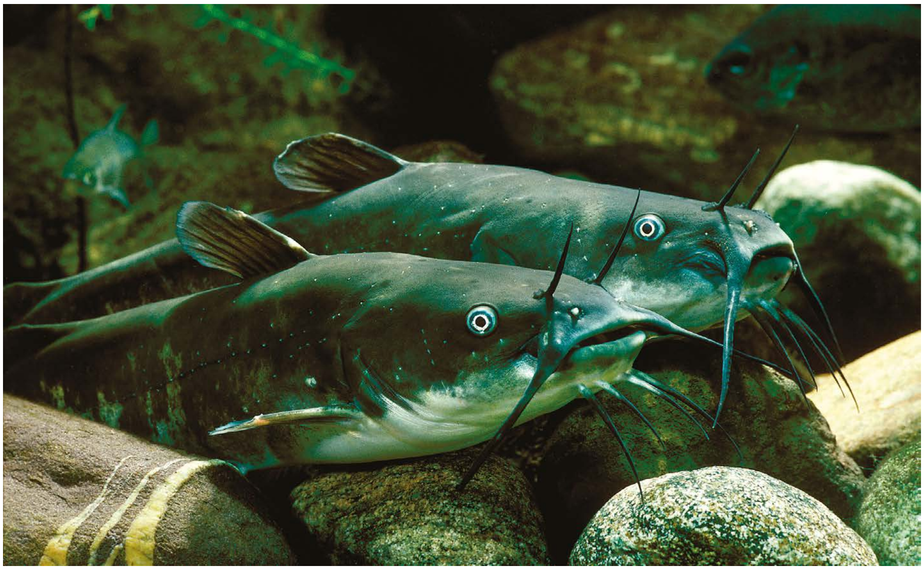

Mysterious black lesions were found on brown bullhead catfish (*Ameiurus nebulosus*).

PHIL DEGGINGER/ALAMY

RARE CONTAGIOUS CANCER DISCOVERED IN WILD CATFISH

Transmissible tumours have been found in only a few types of animal, but there might be more out there.

By Ewen Callaway

Researchers have discovered a contagious cancer in catfish – just the fourth example of tumour cells that spread naturally from one animal to another.

The others are a sexually transmitted tumour in dogs, two distinct cancer lines that threaten Tasmanian devils (*Sarcophilus harrisii*) and numerous independent tumour lineages affecting bivalves, including clams, cockles and mussels.

The latest findings explain the mysterious appearance, more than a decade ago, of black skin lesions in brown bullhead catfish (*Ameiurus nebulosus*) in a lake straddling the Canada–US border – and possibly observations elsewhere that date back to the nineteenth century.

Anglers and wildlife biologists first noticed the tumours in 2012 in fish from Lake Memphremagog, which stretches from Newport in Vermont to Magog, Canada, with later surveys suggesting that about one-third of the lake's brown bullhead catfish were afflicted[1].

The skin lesions, a form of melanoma, are inky black and sometimes crusty, with some fish displaying "tumours on top of tumours", and can spread to the brain, liver and other organs, says Julie Dragon, a data scientist at the University of Vermont in Burlington and the lead author of a study published on 22 July in *Nature*[2]. But the health effects are unclear, because the tumours appear in young and old, mature fish. "We don't have evidence of a huge age cliff that suggests it's killing the fish," Dragon adds.

Fishwives' tale

Michael Metzger, a biologist at the Pacific Northwest Research Institute in Seattle, Washington, isn't surprised that another transmissible cancer has been found, but he wasn't expecting it in a freshwater species. Metzger's group has found that marine-bivalve transmissible tumour cells can survive for days in salt water – supporting the theory that they spread by way of free-floating cells – but die instantly in freshwater[3]. He wonders whether the catfish tumours transmit through direct contact, like those affecting dogs and Tasmanian devils.

Inky black tumours visible near the fish's tail.

VICKI S. BLAZER/USGS

The discovery of yet another contagious cancer is "incredibly exciting" to researchers, says Elizabeth Murchison, a geneticist at the University of Cambridge, UK, who studies the dog and Tasmanian-devil tumours. "It suggests that transmissible cancers are not as rare as people think and there may well be more of them out there." Comparisons between the different cancers could help researchers to understand underlying causes.

Pollution was a suspected cause, as was a viral infection. Working with Vermont state biologists, Dragon found no evidence for any widespread virus. The team cast a wider net and sequenced the genomes of individual fish, in search of potential cancer-causing DNA mutations that were common in the lake's brown bullhead catfish.

Instead, they found hallmarks of a transmissible tumour. Tumour cells from different fish shared hundreds of thousands of gene variants, forming a single genetic lineage that was distinct from healthy cells in the same individual fish. "We had no inkling that this was going to be the case," Dragon says. "It's still mind-blowing."

A key question is whether this is because scientists now have the technologies to easily detect transmissible cancer, or because of wider ecological changes increasing its prevalence, says Andrew Storfer, an evolutionary biologist at Washington State University in Pullman. The dog tumour type evolved more than 6,000 years ago, but the others might have evolved much more recently.

There are also hints that the catfish melanoma could be older and more widespread than the latest data suggest. A 1925 paper documented a similar disease in brown bullhead catfish from a pond in Cape Cod, Massachusetts, and showed it could spread to healthy fish (see go.nature.com/3rtrwvz). Decades earlier, the writer Henry David Thoreau described how half the head of a catfish in a Massachusetts river was "covered with an inky-black kind of leprosy, like a crustaceous lichen".

Dragon says her team has confirmed in unpublished work that the transmissible melanoma is in brown bullhead catfish from other places. "There are actually anglers posting pictures that look very much like this condition."

Additional reporting by Benjamin Thompson.

1. Blazer, V. S., Shaw, C. H., Smith, C. R., Emerson, P. & Jones, T. *J. Fish Dis.* **43**, 91–100 (2020).
2. Curd, E. E. *et al. Nature* https://doi.org/10.1038/s41586-026-10828-6 (2026).
3. Giersch, R. M. *et al. Pathogens* **11**, 283 (2022).